Collaborator or Assistant?

How AI Coding Agents Partition Work Across Pull Request Lifecycles

Agent-associated PRs now enter real projects at scale. What does the visible trace show when they merge?

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Our contribution: a map of where to look

Raw PR metadata records visible actors and events - not hidden authorship, authorization, or accountability. We turn it into a map.

1

Each PR is a lifecycle

A process to characterize — not a row in a table.

2

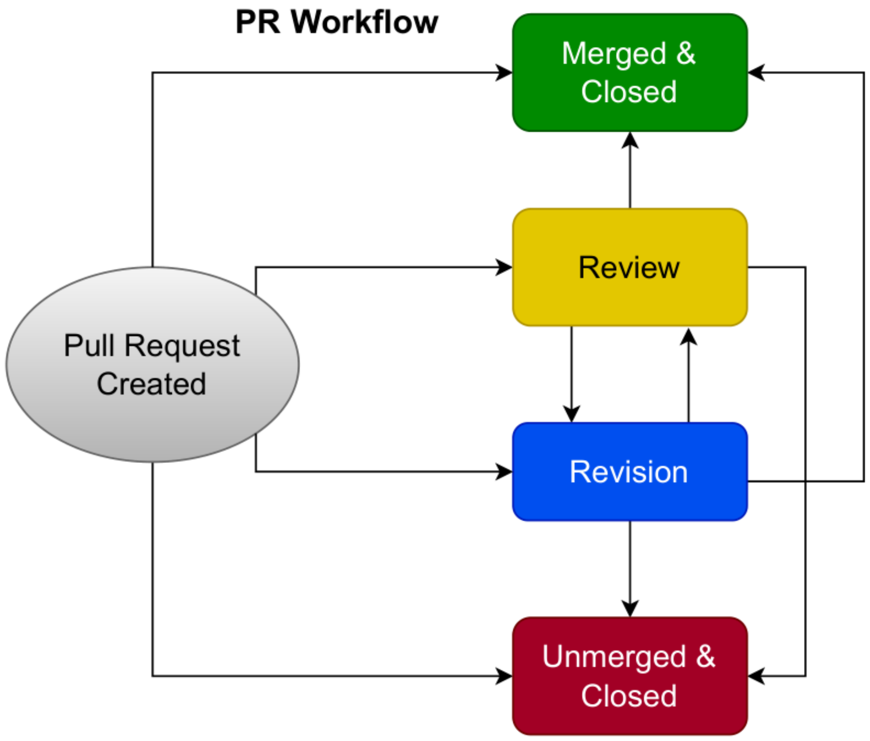
Timeline maps the pathways

State-machine transitions recover the routes a PR can take.

3

Three separable observables

Visible attribution - review trace - merge execution: what surfaces where.



The paper's PR-lifecycle state machine (the map).

Same data, a different lens

AIDev gives an agent-associated corpus; Collaborator or Assistant? maps the visible lifecycle trace inside it. Same PRs, same metadata - a clearer construct boundary.

1. ## AIDev label

   Agent-associated by signal-based curation - not a guarantee of exclusive agent authorship.

2. ## Trace label

   'Human-Init' means human-surfaced first-commit trace, not Human-PR population or verified human authorship.

3. ## Governance finding

   The gap between agent association and human-surfaced attribution is the governance phenomenon.

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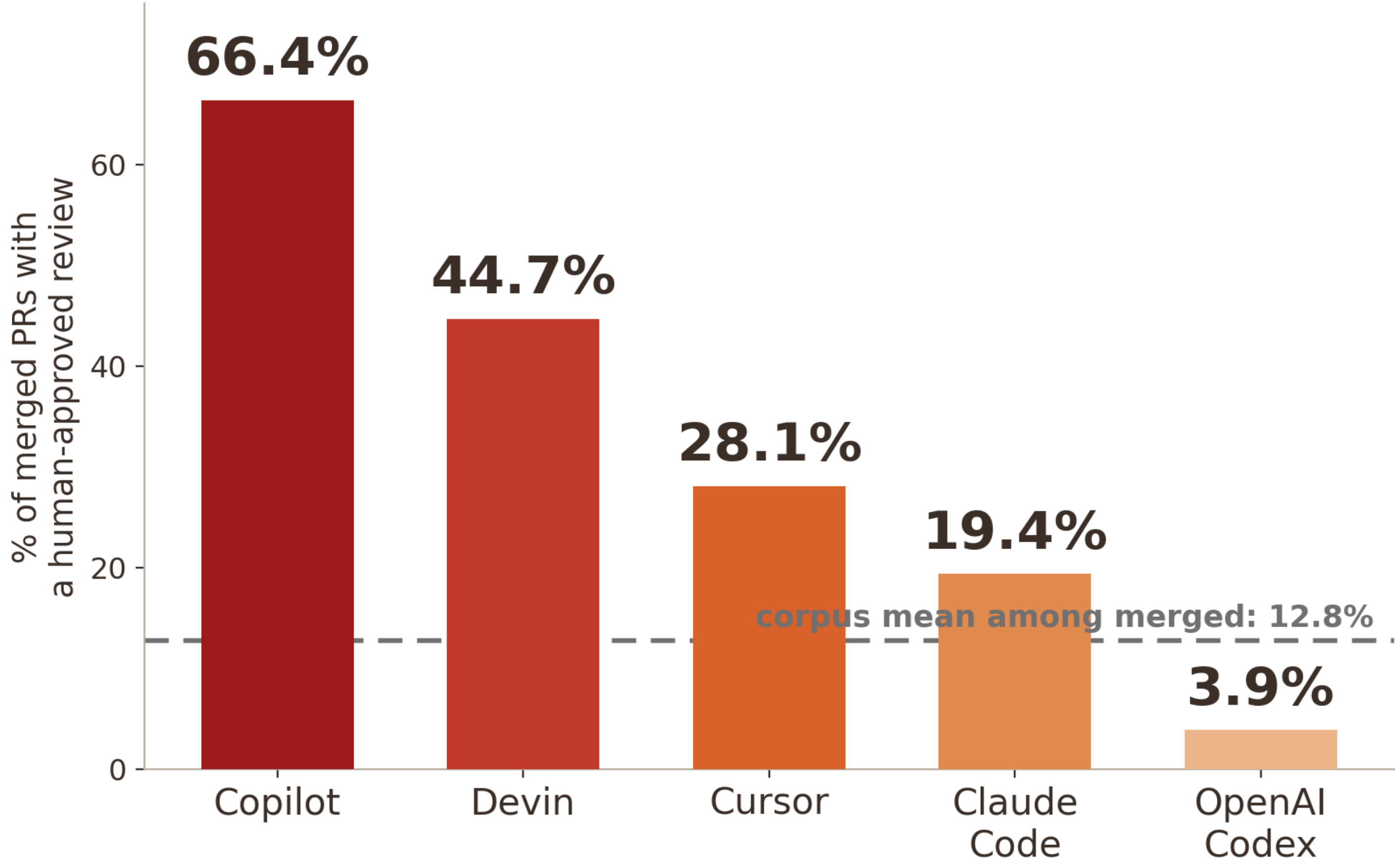
The twist: merge stays visible; review trace often does not



89.6%

of 29,585 terminal PRs lack a recorded human-approved review trace

17×

across the Collaborator→Assistant spectrum (Copilot 66.4% → Codex 3.9%)

Most merged agent-associated PRs lack a recorded human-approved review trace.

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Collaborator or Assistant? See the poster

For visibly agent-attributed tools, the first-commit surface is usually agent-visible; 54 merged PRs are automation-authorized (0.24% of merged PRs), but the trace records merge execution, not necessarily the governance decision. Operational attribution and governance execution decouple.

- **The per-tool governance profile —** behind the 17-fold.
- **The construct correction —** human-surfaced trace, not hidden authorship.
- **Trace vs. governance legibility —** what a recorded merge can and cannot evidence.
- **Downstream-quality check — bounded null, not a harm claim.**

Come to the poster — the instrument, and what it reveals.

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